LAB 9 – NAT, Routing and ACL Troubleshooting

Course: The Bits and Bytes of Computer Networking
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Author: Florencia Horita

1. Introduction

This lab aims to diagnose and solve connectivity issues in a network that uses static routing, NAT, PAT, and Access Control Lists (ACL).

A real-world scenario is simulated where multiple configuration errors prevent communication between a private network and an external server.

The main focus is on analyzing and resolving issues using troubleshooting techniques.

2. Network Topology

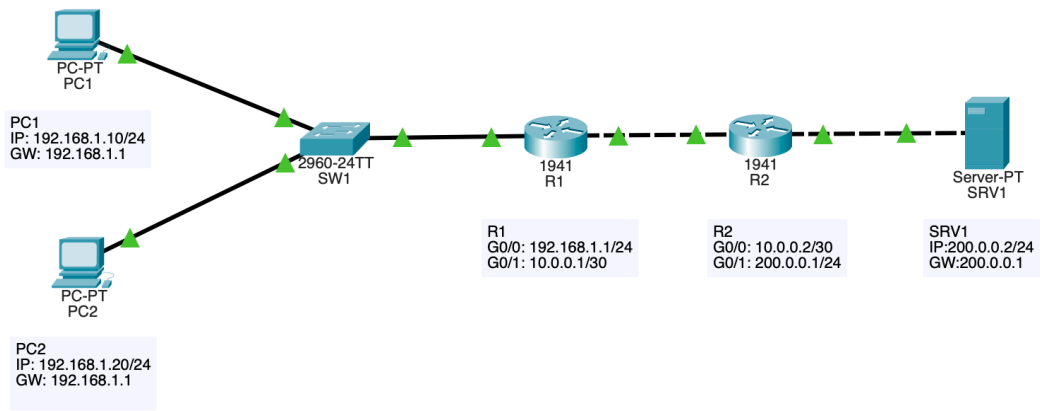
The network used in this lab consists of:

- Two PCs (PC1 and PC2)
- One switch (SW1)
- Two routers (R1 and R2)
- One server

The connection is established as follows:

PC1 / PC2 → SW1 → R1 → R2 → Server

Figure 1 – Network Topology



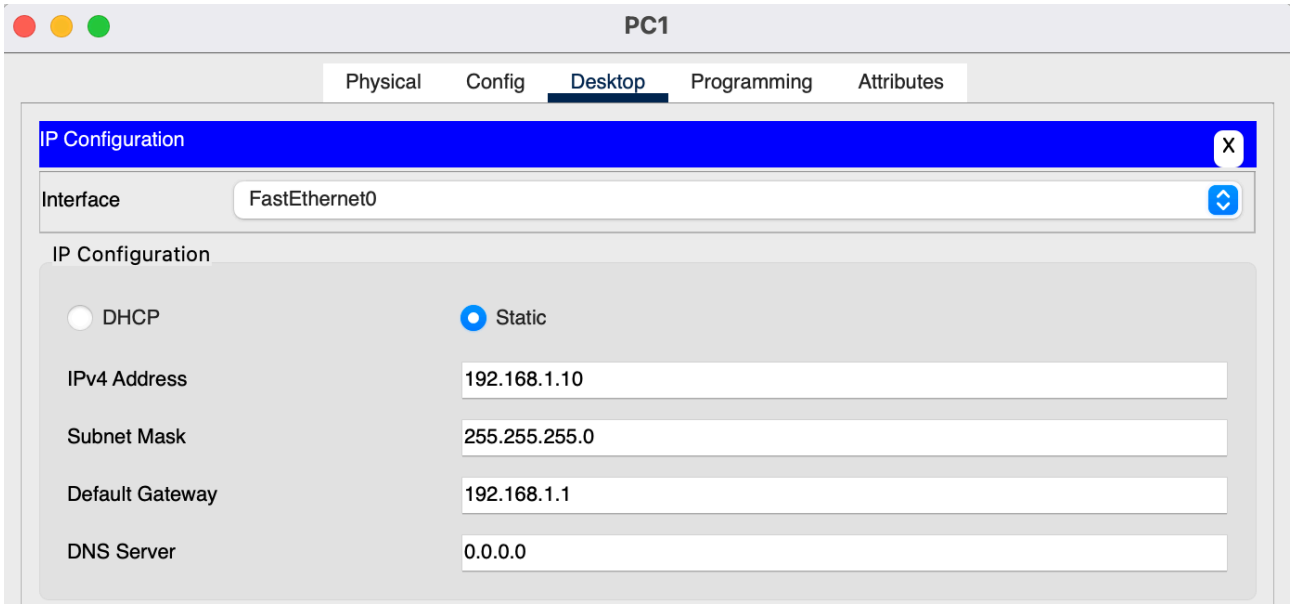
3. Network Configuration

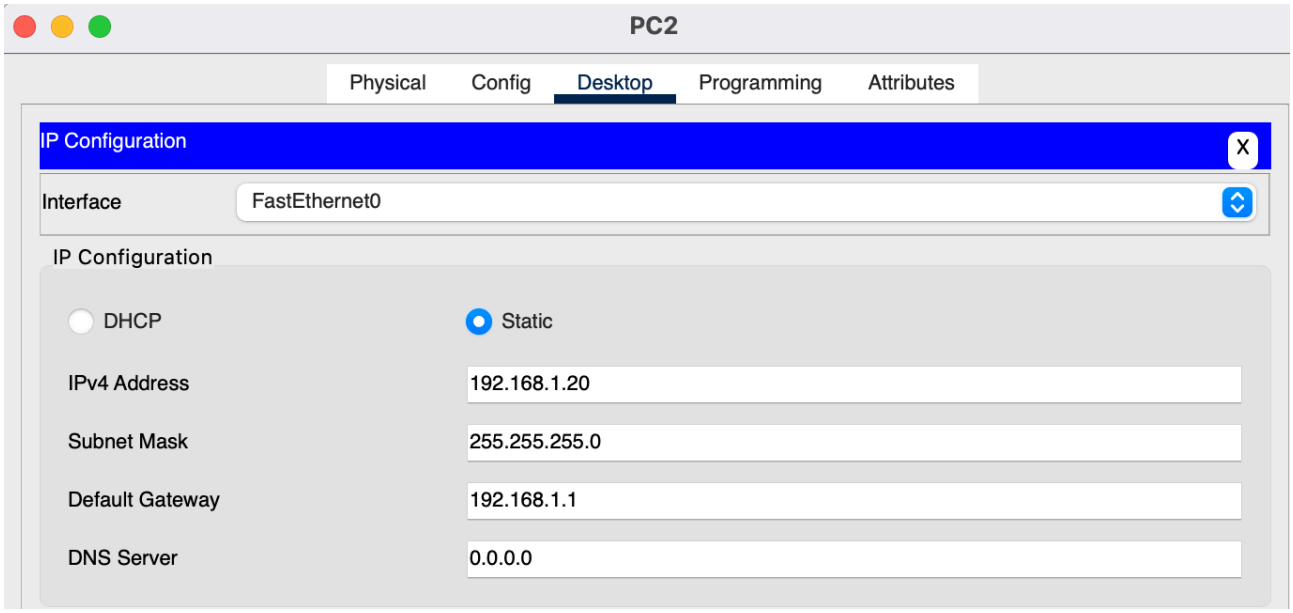
3.1 PC Configuration

PC1
• IP: 192.168.1.10
• Subnet Mask: 255.255.255.0
• Gateway: 192.168.1.1

PC2
• IP: 192.168.1.20
• Subnet Mask: 255.255.255.0
• Gateway: 192.168.1.1

Figure 2 – PC IP Configuration

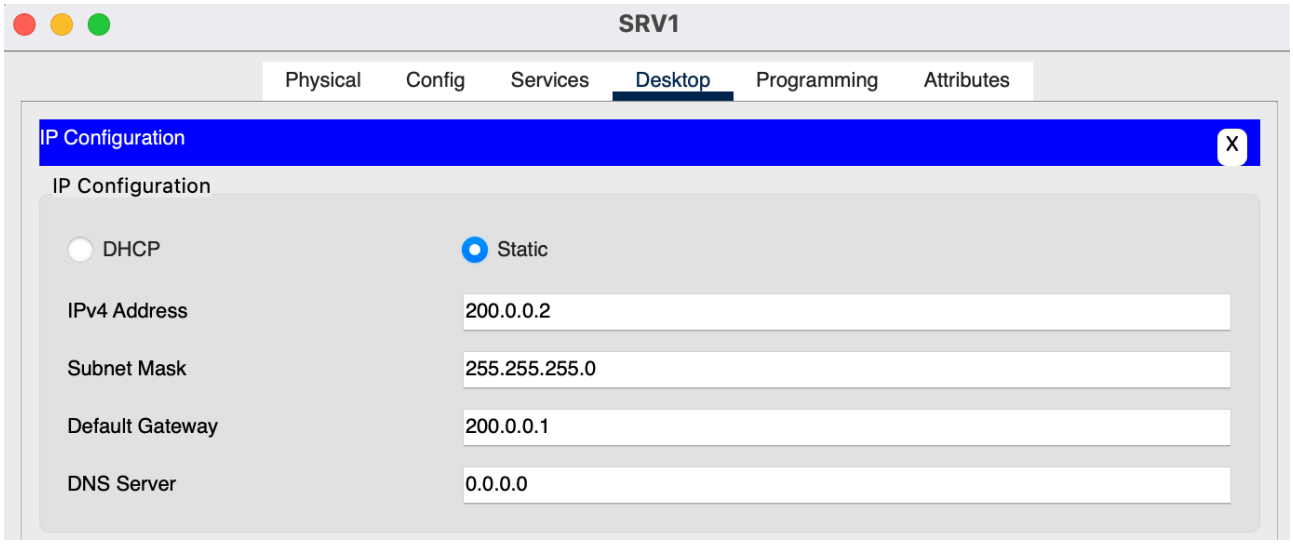




3.2 Server Configuration

• IP: 200.0.0.2
• Subnet Mask: 255.255.255.0
• Gateway: 200.0.0.1

Figure 3 – Server IP Configuration



3.3 Addressing Scheme

Device	Interface	IP Address
PC1	—	192.168.1.10
PC2	—	192.168.1.20
R1	G0/0	192.168.1.1
R1	G0/1	10.0.0.1
R2	G0/0	10.0.0.2
R2	G0/1	200.0.0.1
Server	—	200.0.0.2

4. Routing Configuration

Static routes were configured to enable communication between networks.

4.1 Route on R1

ip route 200.0.0.0 255.255.255.0 10.0.0.2

This route allows R1 to send traffic to the server network.

4.2 Route on R2

ip route 192.168.1.0 255.255.255.0 10.0.0.1

This route allows traffic to return to the internal network.

Figure 4 – Routing table on R1

```
Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
```

Figure 5 – Routing table on R2

```
Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
     200.0.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.0.0.0/24 is directly connected, GigabitEthernet0/1
L       200.0.0.1/32 is directly connected, GigabitEthernet0/1
```

5. NAT and PAT Configuration (R1)

5.1 Internal Network Definition

access-list 1 permit 192.168.1.0 0.0.0.255

Defines which private addresses will be translated.

5.2 Interface Configuration

interface g0/0
ip nat inside

interface g0/1
ip nat outside

- Inside → private network
  - Outside → external network

5.3 PAT Configuration

ip nat inside source list 1 interface g0/1 overload

Allows multiple connections using a single public IP address.

Figure 6 – NAT Configuration

```
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#interface gig0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#interface gig0/1
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#ip nat inside source list 1 interface gig0/1 overload
```

6. Troubleshooting

During the lab, multiple configuration errors were detected.

6.1 Initial Connectivity Test

ping 200.0.0.2

Result: Communication failure

Figure 7 – Failed ping

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 200.0.0.2

Pinging 200.0.0.2 with 32 bytes of data:

Reply from 192.168.1.1: Destination host unreachable.
Request timed out.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.

Ping statistics for 200.0.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

6.2 NAT Verification

show ip nat translations

Result: No NAT translations are observed. This occurs because traffic cannot leave the router, likely due to a routing issue, so NAT is not applied.

Figure 8 – NAT table with no entries

```
Router#show ip nat translations
Router#
```

6.3 Routing Verification

show ip route

Result: Missing or incorrect route

Figure 9 – Routing on R1

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
```

During the initial verification, an incorrect routing configuration is observed, preventing communication. After applying corrections, the routing table shows the necessary entries to reach all networks.

Figure 10 – Routing on R2

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
     200.0.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.0.0.0/24 is directly connected, GigabitEthernet0/1
L       200.0.0.1/32 is directly connected, GigabitEthernet0/1
```

The R2 router does not have a route to the 192.168.1.0/24 network, preventing traffic from returning to the internal network.

6.4 ACL Verification

show access-lists

The configured ACL correctly permits traffic from the 192.168.1.0/24 network, so no issues are detected at this stage.

Figure 10 – ACL Verification

```
Router>enable
Router#show access-lists
Standard IP access list 1
    10 permit 192.168.1.0 0.0.0.255
```

6.5 Interface Verification

show ip interface brief

Result: Interface configuration verified

Figure 11 – Interface status

R1

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     10.0.0.1        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

R2

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.0.2        YES manual up                    up
GigabitEthernet0/1     200.0.0.1       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

All interfaces are in an up/up state, indicating no physical or link-layer issues.

7. Problem Resolution

The following corrections were made:

• Correction of the subnet mask in the router-to-router link (/30)
• Addition of missing static routes in R1 and R2
• Verification of correct NAT configuration (inside/outside)
• Verification of the ACL used for NAT
• Verification of interface status

7.1 Applied Fix – Routing on R1

The absence of a route to the 200.0.0.0/24 network on R1 was detected, preventing traffic from reaching the server.

A static route was configured using 10.0.0.2 as the next hop:

ip route 200.0.0.0 255.255.255.0 10.0.0.2

As a result, R1 can correctly forward traffic to R2.

Figure 12 – Static route added on R1

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
S    200.0.0.0/24 [1/0] via 10.0.0.2
```

8. Final Verification

8.1 Connectivity Test

ping 200.0.0.2

Result:
Communication with the server is established. An initial packet loss is observed, after which connectivity remains stable.

Figure 12 – Successful ping

```
C:\>ping 200.0.0.2

Pinging 200.0.0.2 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 200.0.0.2: bytes=32 time<1ms TTL=126
Reply from 200.0.0.2: bytes=32 time=1ms TTL=126

Ping statistics for 200.0.0.2:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

8.2 NAT Verification

show ip nat translations

Result:
Entries are observed in the NAT table where the private address 192.168.1.10 is translated to 10.0.0.1, confirming correct NAT/PAT operation.

Figure 13 – NAT working

```
Router#show running-config | include ip nat
 ip nat inside
 ip nat outside
ip nat inside source list 1 interface GigabitEthernet0/1 overload
Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 10.0.0.1:13       192.168.1.10:13    200.0.0.2:13       200.0.0.2:13
icmp 10.0.0.1:14       192.168.1.10:14    200.0.0.2:14       200.0.0.2:14
icmp 10.0.0.1:15       192.168.1.10:15    200.0.0.2:15       200.0.0.2:15
icmp 10.0.0.1:16       192.168.1.10:16    200.0.0.2:16       200.0.0.2:16
```

9. Analysis

The communication flow was successfully established after resolving routing issues:

• PCs send traffic to the server
• R1 translates addresses using NAT/PAT
• R2 correctly routes the traffic
• The server responds properly

The following were successfully validated:

• Routing
• NAT
• PAT
• ACL (verified without configuration errors)

10. Conclusion

This lab allowed the application of troubleshooting techniques in a realistic environment, identifying and resolving routing and network configuration errors that prevented connectivity.

Key skills such as fault diagnosis, configuration analysis, and connectivity validation were reinforced, achieving successful communication between all network devices.